

General Purpose Utilities

cal, date, who, whoami, tty, uname, passwd, echo, tput, be,script, wall, write, mail.

1. cal

It displays calendar of prev/current/next month if no argument is specified.

Displays calendar of specific year if specified.

\$ cal

Table 2.

Options	Function
-1	Display single month output.
-3	Display prev/current/next month output.
-s	Display Sunday as first day of week.
-m	Display Monday as first day of week.
-j	Display Julian dates.
-y	Display a calendar for current year.

2. date

Following command displays current date & time if no argument is given.

\$ date

Tue May 6 02:45:40 IST 2003

Formatted output for date command:

Field Descriptor	Function
%a	Abbreviated weekdays(SUN to SAT)
%A	Full weekday name
%b	Abbreviated month name
%B	Full month name
%c	Current date & time
%C	Century (00 to 99)
%d	Day of month 0 to 31
%D	Day as MM/DD/YY
%e	Day of month as decimal number
%h	Abbreviated month
%H	Hour 00 to 23
%I	01 to 12 (12 hour clock)
%j	Day of the year 001 to 366
%m	Month of the year 01 to 12
%M	Minute 00 to 59
%n	New time
%p	AM/PM
%r	Time in AM/PM notation
%T	Time in hh:mm:ss (23 hrs clock)

e.g. To display the date & time in the format

DATE=

TIME=

DAY=

Command is `$ date '+DATE=%D%n TIME=%T%n DAY=%A'`

To use multiple specifiers, enclose them within quotes and use + symbol before it.

`$ date +%m`

05

`$ date +'%h %d'`

May 03

3. **who**

This command displays the information of users who are currently logged in. Execute following command and write the output(only 3 users)

`$ who`

Execute the following options and write the use of it.

4. **who am i**

This command displays the information of the user who is currently logged in.

Execute the following command and write the output:

`$who am i`

5. **Stty (set tty)**

This command allows the user to set, view and change a wide variety of terminal settings.

E.g., end of line character, speed, end of file character, start, stop, suspend character etc.

Observe and the output of the following command:

`$stty`

`$stty-a`

To set terminal settings like erase character to ^e, observe observe the output of the following command

`$stty erase \^e`

6. **Uname**

Uname command is used to display basic information about the operating system and hardware. With options, Uname prints kernel details, and system architecture.

Uname is the short name for 'UNIX name'. Uname command works on all Linux and Unix like operating systems.

When you run Uname command without any options it just prints the core operating system name.

Syntax:

uname [OPTION]

- **-a option:** It prints all the system information in the following order:
Kernel name, network node hostname, kernel release date, kernel version, machine hardware name, hardware platform, operating system
- **-s option:** It prints the kernel name.
- **-n option:** It prints the hostname of the network node(current computer).
- **-r option:** It prints the kernel release date.
- **-v option:** It prints the version of the current kernel.
- **-m option:** It prints the machine hardware name.
- **-p option:** It prints the type of the processor.
- **-i option:** It prints the platform of the hardware.
- **-o option:** It prints the name of the operating system.

Refer link for more information

<https://www.geeksforgeeks.org/uname-command-in-linux-with-examples/>

7. passwd

This command changes the old password of the current user to the new one.

```
$ passwd
```

Changing password for the user pramod.

Changing password for pramod

(Current) UNIX password:

New password:

Retype new password:

8. echo

The following command without specifying its arguments within quotes squeezes tabs to a single space as below.

```
$ echo Hello      Dinesh
```

Hello Dinesh

To preserve spaces, include the arguments within quotes as given below:

```
$ echo "Hello      Dinesh"
```

Hello Dinesh

```
$
```

Note: By default, echo command does not interpret the escape characters like \n, \t when included in the text to display. You can enable their interpretation by -e option, and disable by -E option.

```
$ echo -e "hello\n"
```

```
Hello
```

```
$
```

As shown in above command new line has been inserted after the hello string.

9. tput

It initializes a terminal or query terminfo database

```
tput [-Ttype] capname [parms ... ]
```

```
tput [-Ttype] init
```

```
tput [-Ttype] reset
```

```
tput [-Ttype] longname
```

```
tput -S <<
```

```
tput -V
```

The **tput** utility uses the **terminfo** database to make the values of terminal-dependent capabilities and information available to the shell (see **sh(1)**), to initialize or reset the terminal, or return the long name of the requested terminal type. The result depends upon the capability's type:

Tag	Description
string	
	tput writes the string to the standard output. No trailing newline is supplied.
integer	
	tput writes the decimal value to the standard output, with a trailing newline.
boolean	
	tput simply sets the exit code (0 for TRUE if the terminal has the capability, 1 for FALSE if it does not), and writes nothing to the standard output.

Refer link for more information

https://www.tutorialspoint.com/unix_commands/tput.htm

10. Script

Script makes a typescript of everything printed on your terminal. It is useful for students who need a hardcopy record of an interactive session as proof of an assignment, as the typescript file can be printed out later with `lpr(1)`.

If the argument file is given, script saves all dialogue in file. If no file name is given, the typescript is saved in the file `typescript`.

Options:

Tag	Description
-a	Append the output to <i>file</i> or typescript, retaining the prior contents.
-c <i>COMMAND</i>	
	Run the <i>COMMAND</i> rather than an interactive shell. This makes it easy for a script to capture the output of a program that behaves differently when its stdout is not a tty.
-f	Flush output after each write. This is nice for telecooperation: One person does ‘ <code>mkfifo foo; script -f foo</code> ’ and another can supervise real-time what is being done using ‘ <code>cat foo</code> ’.
-q	Be quiet.
-t	Output timing data to standard error. This data contains two fields, separated by a space. The first field indicates how much time elapsed since the previous output. The second field indicates how many characters were output this time. This information can be used to replay typescripts with realistic typing and output delays.

11. wall command

`wall` command is used to send messages to all the users currently logged on to the system.

Example: If the superuser logged on `tty00` wishes to send messages to all users.

Syntax:

```
#wall
```

```
_____ # text message
```

```
_____
```

```
_____
```

```
^d
```

```
#
```

Receivers of the message see the following on the screen

Broadcast messages from the root console on `tty00` Aug 1

```
_____ # text message
```

```
_____
```

```
<eof>
```

12. write command

The write command enables communication between users logged on to the system. The command sends a message to a specified user at the time the message is typed, provided the recipient user is logged on to system. Messages can be sent to and fro till the command is terminated using end of text character ^D on a line by itself. For two users to communicate to each other, each of them has to invoke write command.

Syntax:

```
$ write userid [terminal]
```

```
_____# text of message
```

```
_____
```

```
^d      # end of message
```

```
$
```

13. Talk command

The talk command can be used in the place of write for two way communication. The difference is that the viewers screen is divided into two halves where sender's screen is displayed as upper half and receiver's screen as lower half for both sender/receiver [chat facility on internet].

14. mail command

The mail command of UNIX/Linux facilitates the users to send messages to each other. The messages are stored in a mailbox in the /usr/mail directory of the system.

..

..

..

Syntax:**Sending mail**

```
$ mail userid [terminal]
Sub:
_____
_____} text of message
^d
$
```

Reading mail

```
$ mail [options]
{
  Most recent text message
}
&          # mail prompt
Commands at mail prompt
?          help summary
q          quit
x          exit without changing mail
p          print
d          delete
+          print previous
m user     forward mail to user
s [file]   save with header (default mbox)
w[file]    save without header
```